

Biasing the Transistor

To operate as an amplifier a transistor must be **biased** in the active region. In that case, the transistor can amplify a *small ac signal* faithfully.

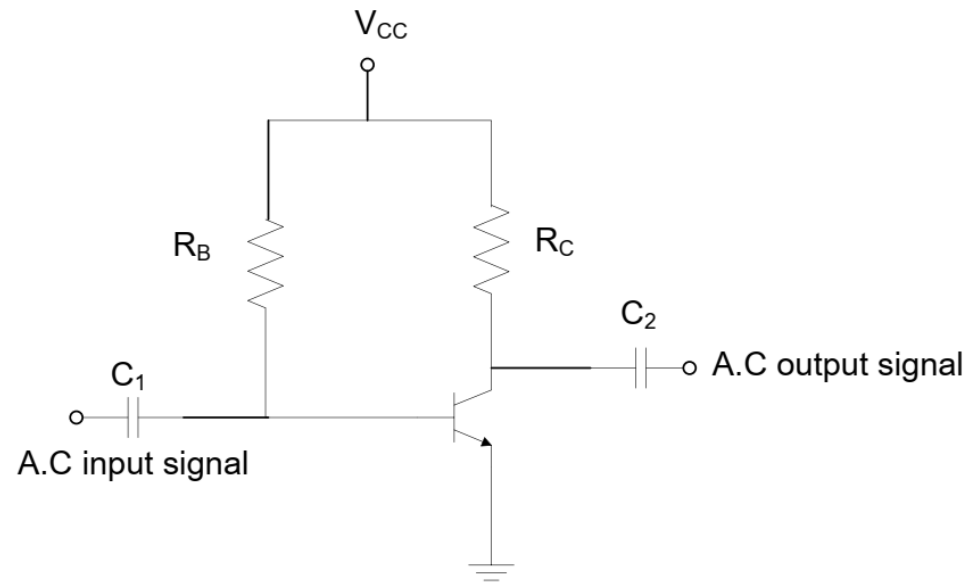
Bias Point (or Quiescent Operating Point, i.e. **Q-Point**) is decided by the DC voltages applied and the values of the resistances used. (See earlier example.)

The Bias Point is specified by the three quantities **V_{CE} , I_C and I_B** . *This point must be in the active region of the transistor if we want to use the transistor (with that bias point) as an amplifier.*

The selected point should be able to accommodate the “output signal swing” without distortion.

Ideally, the biasing should be done in such a way that the Q-point does not change even if there are some changes in β or in the temperature at which the device is operating

Transistor Biasing - Fixed Bias

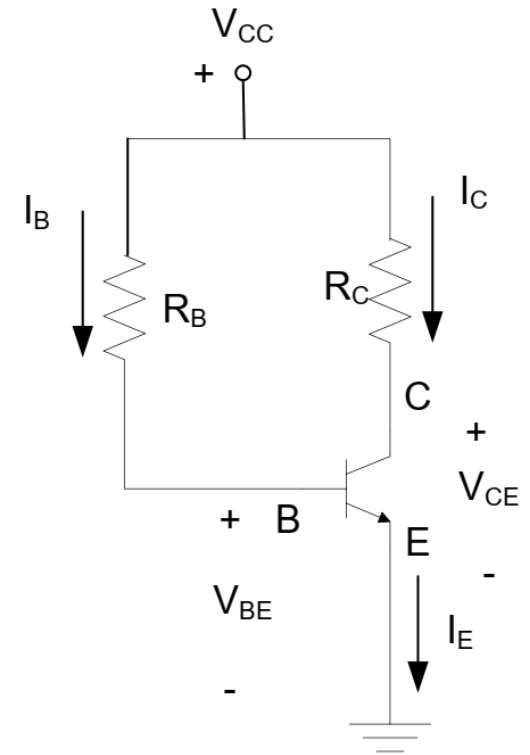


$$I_B = \frac{V_{CC} - 0.7}{R_B}$$

Assuming transistor in active region

$$I_C = \beta I_B = \frac{\beta(V_{CC} - 0.7)}{R_B}$$

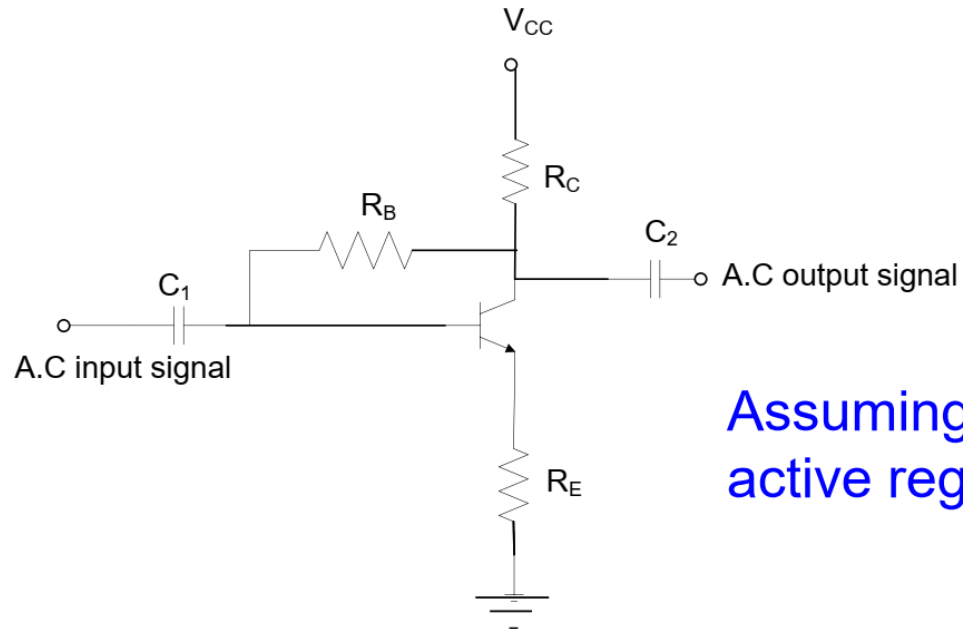
$$V_{CE} = V_{CC} - I_C R_C = V_{CC} - \beta \frac{R_C}{R_B} (V_{CC} - 0.7)$$



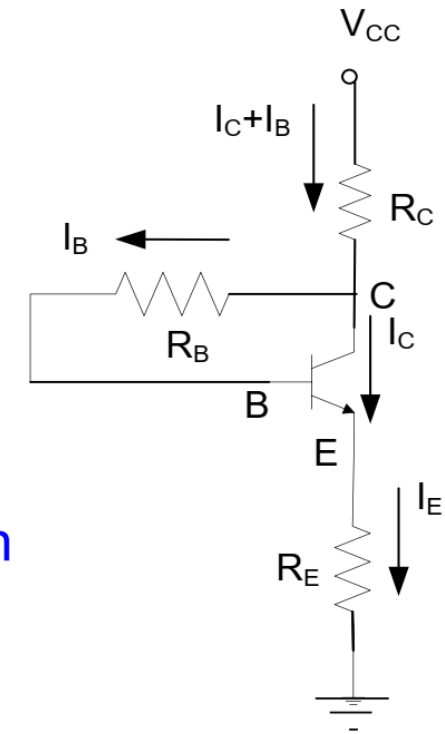
DC Equivalent

C_1 and C_2 will be open circuit under DC

Emitter and Collector Feedback Bias



Assuming transistor in active region



DC Equivalent

$$V_{CC} = (I_C + I_B)R_C + I_B R_B + V_{BE} + (\beta + 1)I_B R_E$$

Using $I_C = \beta I_B$, we get

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)(R_C + R_E)}$$

and

$$V_{CE} = V_{CC} - (I_C + I_B)(R_C + R_E)$$

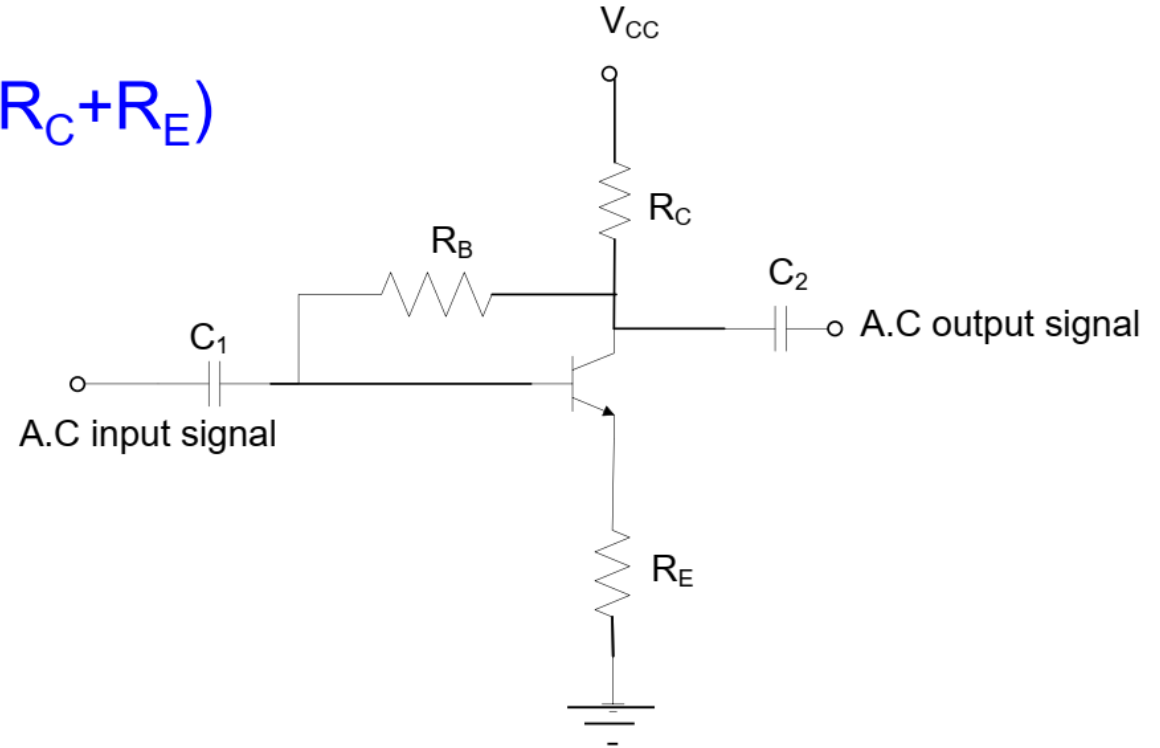
Q-Point

Emitter and Collector Feedback Bias

Typically, β is large so $R_B \ll (\beta + 1)(R_C + R_E)$

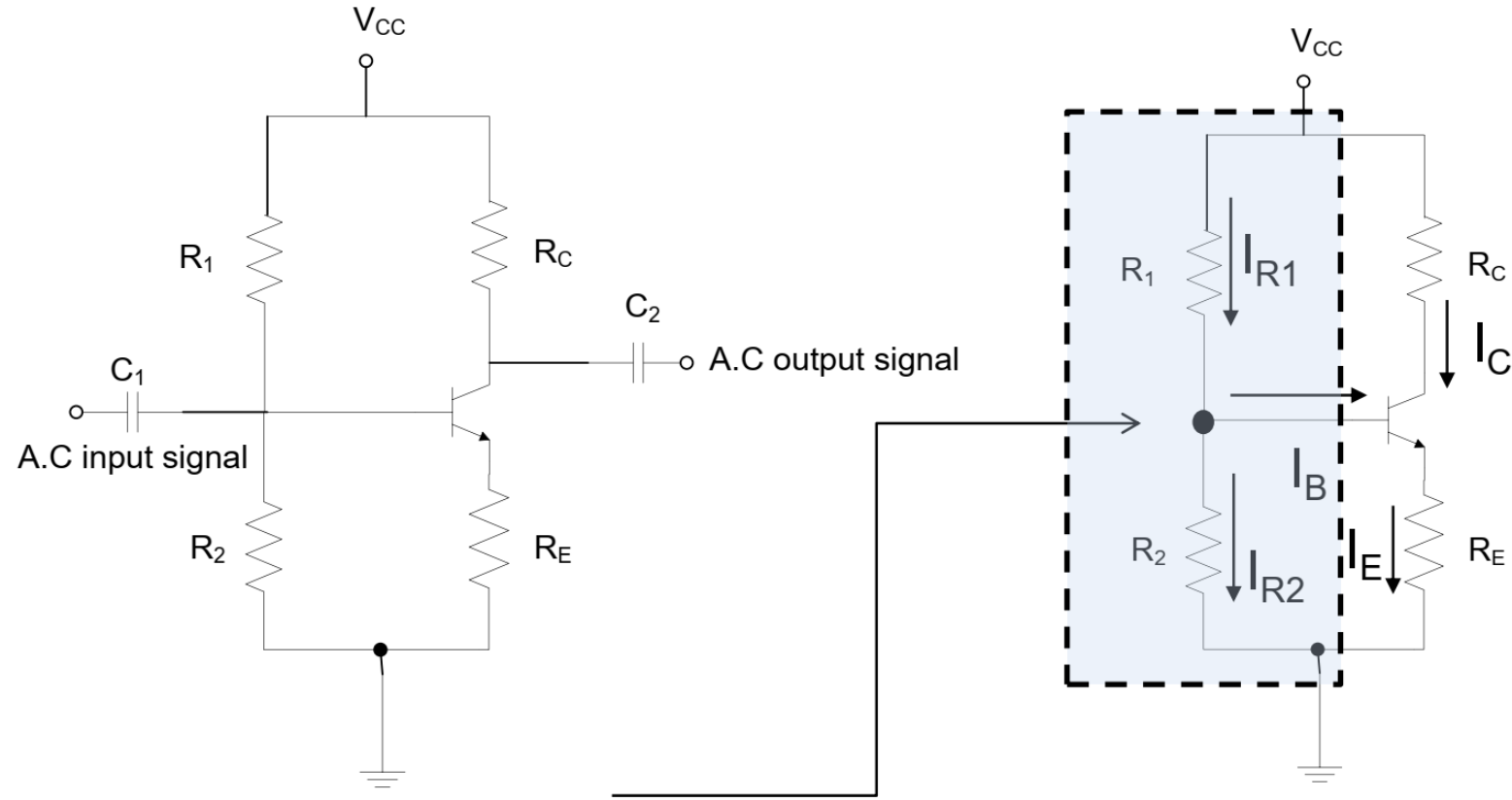
Therefore
$$I_B = \frac{V_{CC} - V_{BE}}{(\beta + 1)(R_C + R_E)}$$

and
$$I_C = \beta I_B \cong \left(\frac{V_{CC} - V_{BE}}{R_C + R_E} \right)$$



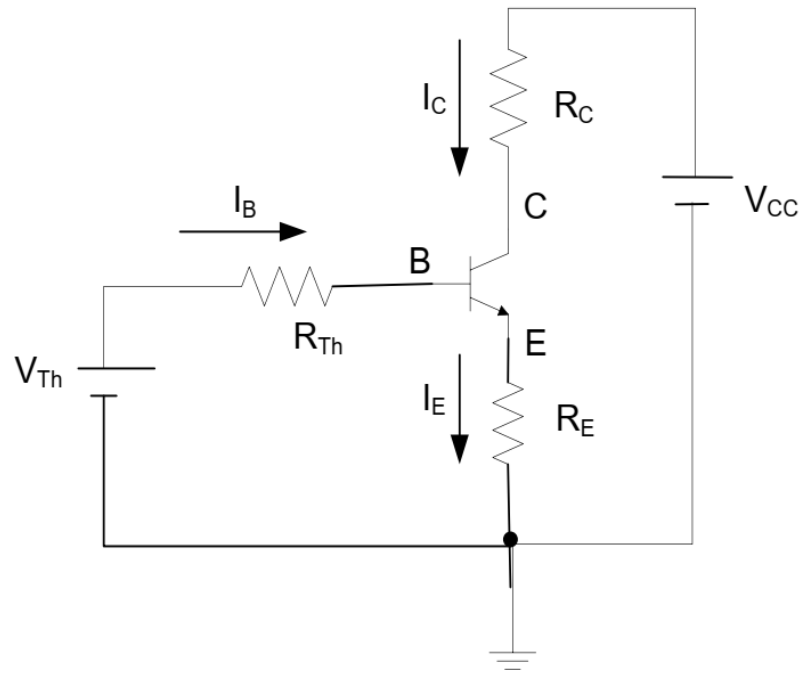
Note the desirable feature that I_C is independent of β as long as we choose a transistor which has a large β . This is an example of *Bias Stabilization* – making the bias point insensitive to β .

Voltage Divider Bias



It is convenient to replace the biasing circuit at the base by its Thevenin's Equivalent using - **DC Equivalent**

$$V_{Th} = V_{CC} \frac{R_2}{R_1 + R_2} = V_{BB} \quad R_{Th} = R_B = \frac{R_1 R_2}{R_1 + R_2}$$



$$V_{Th} = V_{CC} \frac{R_2}{R_1 + R_2} \quad R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

$$V_{Th} = I_B R_{Th} + V_{BE} + (\beta + 1) I_B R_E$$

$$V_{CE} = V_{CC} - I_C R_C - I_E R_E$$

$$\text{with } I_C = \beta I_B \text{ and } I_E = (\beta + 1) I_B$$

For large β , I_C independent of β

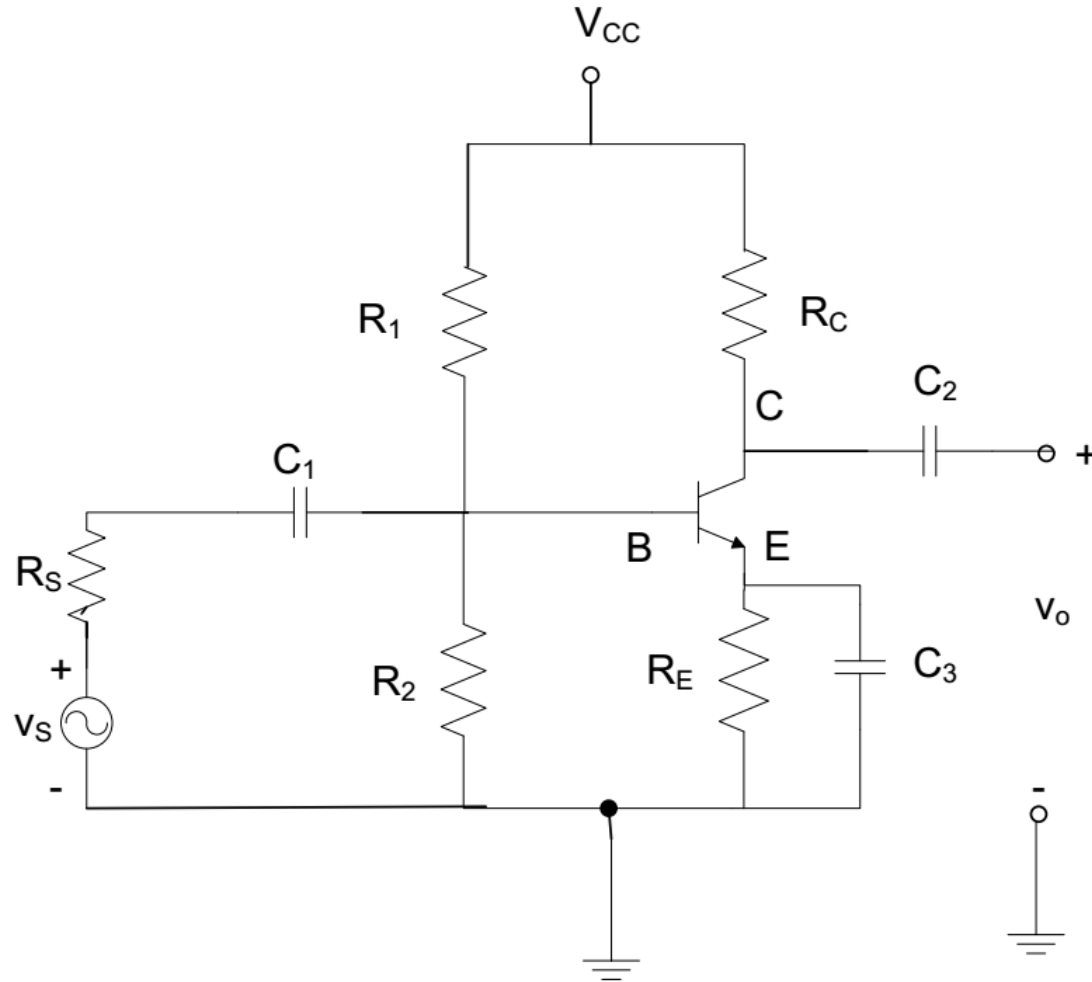
$$I_B = \frac{V_{Th} - V_{BE}}{(\beta + 1) R_E}$$

$$I_C = \frac{V_{Th} - V_{BE}}{R_E}$$

We choose $R_{Th} \ll (\beta + 1) I_B$

This biasing also provides bias stabilization against changes in β . It also provides stabilization against changes in temperature

Basic BJT Amplifier



C_1, C_2 - Coupling Capacitors

C_3 - Emitter Bypass Capacitor

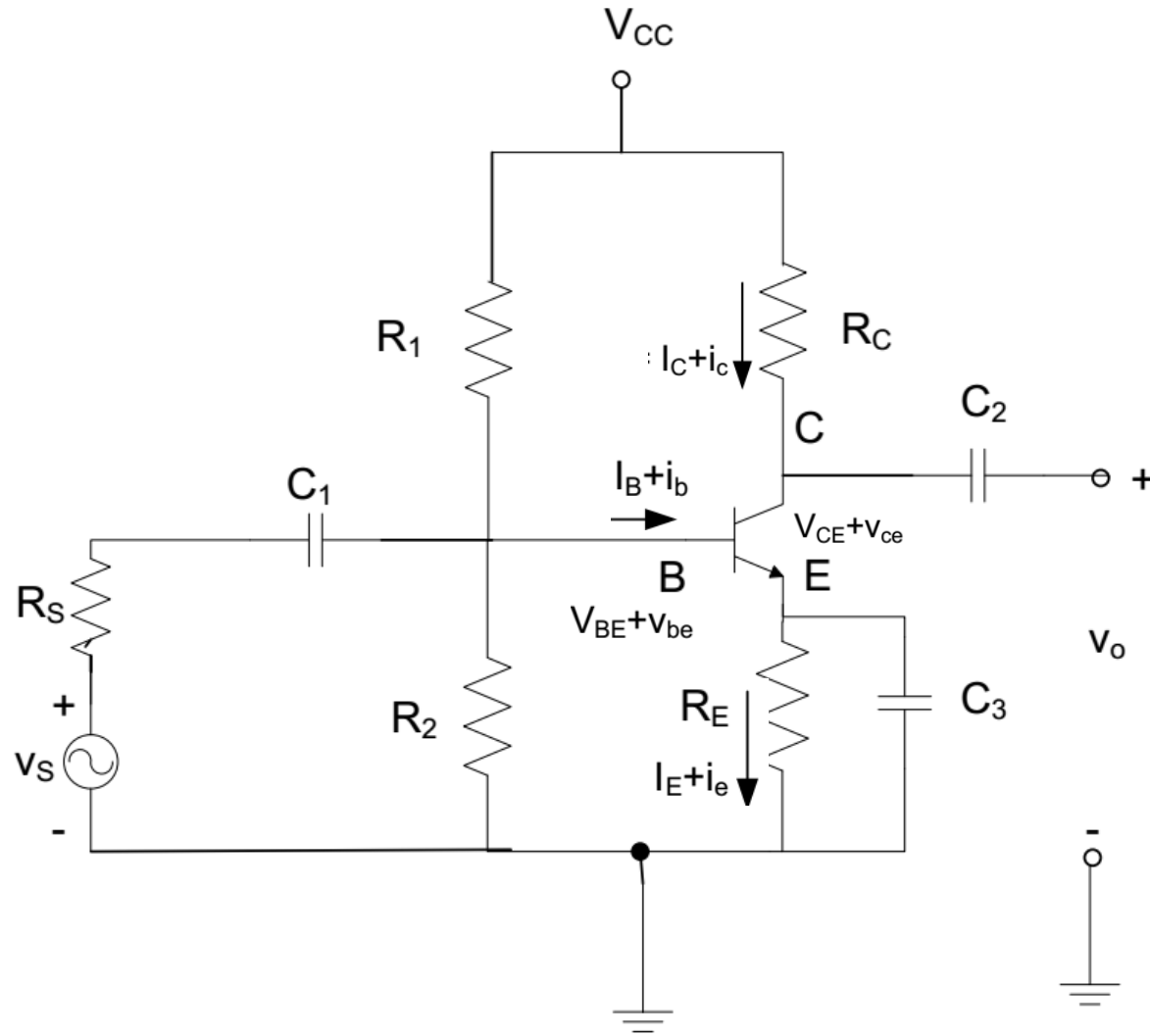
C_1, C_2, C_3 values are chosen high enough so that under ac these act as a short circuit.

The coupling capacitor (C_1, C_2) is used to pass the ac input signal and block the dc voltage from the preceding circuit. (AC-Coupled Amplifier)

This prevents dc in the circuitry on the left of the coupling capacitor from affecting the bias.

The coupling capacitor also blocks the bias of the transistor from affecting the input signal source.

The emitter bypass capacitor (C_3) is used to bypass the R_E and short circuits the ac signal through C_3 since voltage gain decreases because of presence of R_E



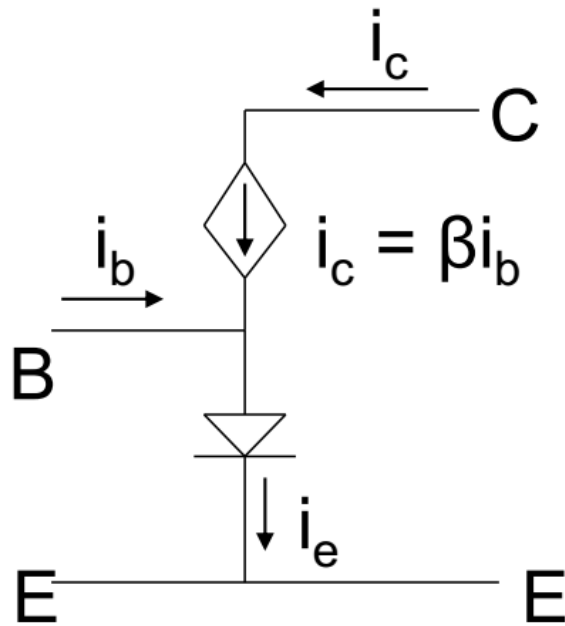
I_B, I_C, I_E – DC currents

i_b, i_c, i_e – AC currents

V_{BE}, V_{CE} – DC Voltages

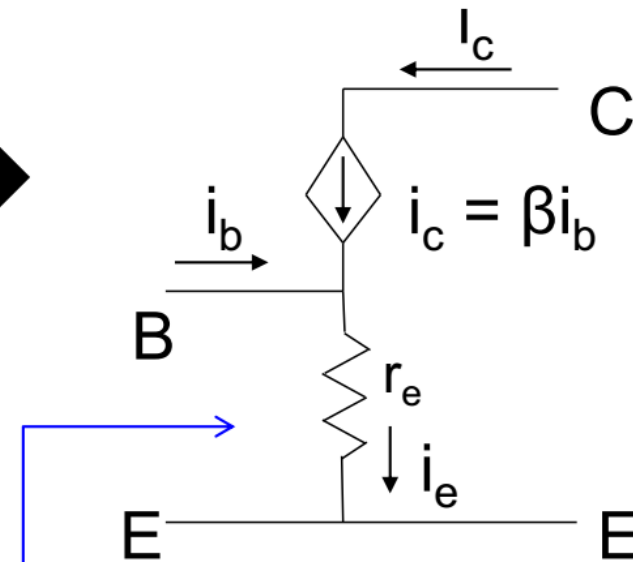
v_{be}, v_{ce} – AC Voltages

**Small Signal AC Equivalent
Model for an NPN Transistor
used in the Common Emitter
Configuration**



$$r_e = \frac{V_T}{I_E} = \frac{26 \text{ mV}}{I_E}$$

I_E is the DC Emitter
Current



Replace BE diode by its
equivalent resistance r_e

Derivation of r_e

$$I_E + i_e = I_{EO} \left(e^{\frac{V_{BE} + v_{be}}{V_T}} - 1 \right)$$

$$r_e = \frac{v_{be}}{i_e}$$

$$\approx I_{EO} e^{\frac{V_{BE} + v_{be}}{V_T}}$$

$$= \frac{V_T}{I_E}$$

$$\approx I_{EO} e^{\frac{V_{BE}}{V_T}} \left(1 + \frac{v_{be}}{V_T} \right)$$

$$i_e = \frac{v_{be}}{V_T} I_E$$

$$= \frac{26mV}{I_E}$$